

San Diego Space Institute Completes Internal Design Review for Sentinel Bio+ Satellite Program

December 24, 2025

The San Diego Space Institute (SDSI) has completed an internal Preliminary Design Review (PDR) for its Sentinel Bio+ (SB+) satellite program, marking a key milestone in the development of a regional space-radiation monitoring infrastructure built largely by university students under professional supervision.

The internal review assessed whether the system architecture, payload design, and spacecraft bus had reached sufficient maturity to proceed toward an external Preliminary Design Review planned for early 2026. The outcome: proceed, with defined action items and no blocking deficiencies.

Sentinel Bio+ is a planned constellation of small satellites designed to measure radiation and space-weather conditions in low Earth orbit. The data will support satellite operators, insurers, and government agencies seeking to better understand and manage the operational risks posed by solar storms and the near-Earth radiation environment.

“This review wasn’t about selling a concept,” said Sean Casey, president of SDSI. “It was about whether the engineering holds together under scrutiny, and whether the teams are ready to take the next step toward flight hardware.”

A Program Built Like Real Infrastructure

Unlike many student satellite efforts, Sentinel Bio+ is structured around industry-style workflows. The internal PDR followed NASA-analog review criteria, including system architecture stability, interface definition, power and mass margins, risk identification, and early evidence of integration feasibility.

Student teams from San Diego State University (payload development) and San José State University (spacecraft bus development) presented their designs, test results, and open risks to SDSI leadership, faculty mentors, and external advisors.

The payload team demonstrated a radiation-measurement system centered on a dedicated controller board that isolates scientific instruments from the spacecraft bus. Early end-to-end data flow testing, using hardware emulators in place of final flight units, was cited as a strength of the presentation.

The spacecraft bus team presented a complete CubeSat architecture covering power, communications, onboard computing, structure, and fault-management concepts. Reviewers emphasized that while the architecture is sound, additional empirical testing and analysis will be required before an external review.

Clear Action Items, No Surprises

The review concluded with a consolidated set of action items to be completed before the external PDR. These include closing interface control documents, completing communications link margin analyses, baselining flight-qualified onboard computing hardware, and demonstrating powered subsystem bring-up.

Importantly, reviewers noted that the remaining gaps are evidentiary rather than conceptual.

“The design isn’t missing pieces,” Casey said. “What’s left is the work you expect at this stage: analysis, testing, and tightening the interfaces.”

SDSI assessed the program as approximately 75 to 80 percent aligned with NASA-level PDR expectations—an intentional position for an internal review conducted prior to final hardware selection.

Workforce Development With Consequences

For SDSI, the review also serves a broader goal: building a regional workforce with direct experience in flight-qualified systems engineering.

Students participating in Sentinel Bio+ are exposed to formal design reviews, requirements tracking, risk management, verification planning, and documentation standards that mirror those used in government and commercial aerospace programs.

“This isn’t a class project,” Casey said. “These students are working inside a real engineering program with real consequences. That experience matters to employers.”

Looking Ahead

With the internal PDR complete, the Sentinel Bio+ team will focus on closing action items and preparing for an external Preliminary Design Review in February 2026. That review will determine readiness to proceed into detailed design and hardware procurement.

The longer-term vision is a deployed constellation providing continuous radiation data while training the engineers who will build and operate the next generation of space infrastructure—many of them, SDSI hopes, right here in San Diego.

No hype. Just disciplined engineering, reviewed early and often.